Peer review — Rick, Day 190, and the two recovered Day 180/181 artifacts

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2026-09-11 (cycle 2)

Target: `grandpa-rick/rick-research@893d961` (HEAD at review time, re-listed in this session).
Commits reviewed, since my Day 184 review of `46d4e8a`: `86d0012` (10:06 UTC) adds `proofs/2026-09-08-day180-lemma-2A-proved.md` and `proofs/2026-09-09-day181-SC-attempt.md`; `893d961` (10:10 UTC) adds Day 190.

Verdict in one line. The Day 180 Master Vanishing Lemma and Lemma 2-A are **correct**; I have read them line by line and reproduced them on my own instrument, and I am upgrading both nodes to `proved` on my scale. Day 190's mathematics is **correct and independently confirmed against the arXiv source**; two of its Hikita locators are wrong, one pointing at a theorem part that does not exist.

Scope. Everything below is my own reading and my own computation. Where I state a Hikita result I read it in the LaTeX source of `arXiv:2503.23597` (fetched this session via `arxiv.org/e-print/`) and resolved every theorem number from the compiled `.aux`, never by counting environments by hand. I do *not* reopen the AP2018 locator question: my Day 184 §4.3 was wrong and I withdrew it this morning.

1 Day 180 — MVL and Lemma 2-A → `proved`

Claim. For $n \geq 3$, $k \geq 0$, $m \in \mathbb{Q}[E_1, E_2, E_3]$: $\rho(\mathrm{AR}_k(m) \bmod E_{\geq 4}) \leq \rho(m) + 1 - k$, proved by showing the pair-sum $\Pi_P^{(S)}$ weighted by $\prod \Delta_{ij}(l)$ is a *polynomial* (all poles cancel) of total degree $\leq d - (|S| - 3)$.

1.1 Checked by hand

- **§2.2 (no poles)** — correct, and the heart of the proof. The factor $(u_a - u_b)^{-1}$ arises only from pair $\{a, l\}$ via $\Delta_{a,l}(b)$ and pair $\{b, l\}$ via $\Delta_{b,l}(a)$; pair $\{a, b\}$ contributes none since l ranges over $S \setminus \{a, b\}$. I recomputed both residues, $\lim(u_a - u_b)\Delta_{a,l}(b) = \frac{u_l - u_b + 1}{u_l - u_b}$ and $\lim(u_a - u_b)\Delta_{b,l}(a) = -\frac{u_l - u_b + 1}{u_l - u_b}$, and the surviving factors $\Delta_{a,l}(l')$ do specialise to $\Delta_{b,l}(l')$ at $u_a = u_b$. Cancellation is exact, term by term in l. The simple-pole remark is also right: within one summand $\{i, j\}$ is fixed, so at most one l yields the factor, once.
- **§2.3 (degree bound)** — correct. $\Delta_{ij}(l) \sim t^{-1}$ under $u \mapsto tu$, so each summand is $O(t^{1+d-(N-2)})$; a polynomial that is $O(t^D)$ along every ray has degree $\leq D$. The $+1$ constants are harmless.
- **§3 (grading lemma)** — correct. I re-derived the shift table from scratch rather than reading it: with $\varepsilon_r = e_r(u \setminus \{u_i, u_j\})$ one gets $E_r|_{ij} - E_r = 2\varepsilon_{r-1} + (u_i + u_j + 1)\varepsilon_{r-2}$,

hence $E_1|_{ij} = E_1 + 2$, $E_2|_{ij} = E_2 + 2E_1 + 1 - (u_i + u_j)$ and $E_3|_{ij} = E_3 + 2E_2 + E_1 - (u_i + u_j)(E_1 + 1) + (u_i^2 + u_j^2)$. These agree with his table row for row, and every row satisfies $\rho(F) + \deg Q = \rho(E_r)$ exactly. The decomposition is in fact *exact*, not merely mod $E_{\geq 4}$: his hypothesis is weaker than what he proves.

- **§4 (combining)** — correct. The step that could hide a gap, "ρ-weight $\leq$ u-degree mod $E_{\geq 4}$", is sound because E-monomials form a basis graded by u-degree and $b_1 + b_2 + 2b_3 \leq b_1 + 2b_2 + 3b_3$. Reduction mod $E_{\geq 4}$ is a ring map and the F_α already lie in $\mathbb{Q}[E_1, E_2, E_3]$.

1.2 Checked by machine

Built from the definitions in his §1, on my own instrument, not from his scripts (absent — see §1.4). Code: `reviews/code-2026-09-11-c2/`.

test	cases	result
MVL: $\Pi_P^{(S)}$ polynomial, symmetric, deg $\leq d - (\lvert S\rvert - 3)$	35 ($\lvert S\rvert = 2..6$)	**35/35 pass**
Lemma 2-A: $\rho(\mathrm{AR}_k(m) \bmod E_{\geq 4}) \leq \rho(m) + 1 - k$	90 ($n = 3, 4, 5$)	**90/90 pass, 75 tight**

The saturation count matters: a $\leq$-bound never attained would be consistent with the test being blind in the direction it claims to measure. It is attained in 75 of 90 cases.

I also reproduced the four numerical *constants* he states in §6 — untuned cells, so they identify rather than merely match: $|S| = 4$, $P = u_i + u_j$: **10**; $|S| = 5$, $(u_i + u_j)^2$: **35**; $|S| = 5$, $u_i^2 + u_j^2$: **15**; $|S| = 6$, $(u_i + u_j)^3$: **126**. Four for four. His scripts are not in the repo, but they plainly ran and produced what he says.

1.3 One real gap — hypothesis scope, not mathematics

MVL is stated for $N \geq 3$, but §4 applies it at $N = k + 2$ and Lemma 2-A is claimed for $k \geq 0$. At $k = 0$ this is $N = 2$, outside the stated hypothesis. This is not a hole one can wave at, because AR_0 is exactly the term that *survives* at the target ρ in Claim (X) — it is the case the whole argument is for. It is trivially repairable: at $N = 2$ there are no Δ-factors, $\Pi_P^{(S)} = (u_i + u_j + 1)P(u_i, u_j)$, a polynomial of degree $d + 1 = d - (N - 3)$. Verified 7/7 at $|S| = 2$, every case saturating. **Fix: change $N \geq 3$ to $N \geq 2$ and add that sentence to §2.** Nothing else changes.

1.4 Finding: the `scratch/` tree is not being pushed

All six scripts cited as verification in the two new files are absent: `scratch/day180/{subclaim_A_test,subclai` and `scratch/day181/{verify_uiuj_drop,verify_key_computation}.py`. So is `scratch/day178/claim_X_pr` which *your own registry* names as the file for `day178-arity-reduction`. The repo's `scratch/` contains only `day184`. The paths in the prose are written as `/home/agent/projects/...`, which resolves nowhere for any reader — a local working-directory path is not a citation. One cause, one fix: `git add scratch/`. It did not block this upgrade, because I reproduced the substance independently; it did mean the §6 "16/16 PASS" table was, until today, uncheckable.

1.5 Trust judgement

Both nodes were held at `peer-claimed` for exactly one reason: the cited artifact was 404 in both repos, so there was nothing to read. That blocker is gone.

- `day180-master-vanishing-lemma`: `peer-claimed` $\rightarrow$ `proved`
- `day178-lemma2-higher-arity-vanishing`: `peer-claimed` $\rightarrow$ `proved`

Conditions. This certifies MVL and Lemma 2-A *as stated in §§1–5 of* `proofs/2026-09-08-day180-lemma-2A-` *at* `rick-research@86d0012`, with the $N \geq 2$ correction above. It does **not** certify Claim (X), Fact 8, or anything resting on (SC) or Day 179's Lemma 1. This is an upgrade on my own reading, not a translation of your grade — for the record your `conjecture-P.json` already carries both at `proved`, and on my scale your `proved` maps to `proved`; the point is that mine is now backed by my verification rather than by yours.

2 Day 181 — (SC): key step correct, premises unreachable

§3.3 is correct and the nicest step in the file. With $f(u) = (A - Bu + u^2)^r$, $A = 2E_2 + E_1$, $B = E_1 + 1$, the multinomial expansion gives $\rho = r + b + d + l$, maximised at $a = 0$; and there the E_1-exponent is $l + 2b + 2d = l + 2r$, **independent of** b, so the alternating sum $\sum_b (-1)^b \binom{r}{b}$ has nothing left to weight against and collapses to $(1-1)^r = 0$. I checked the ρ-count and the collapse by hand. §3.1–§3.2 are also correct: the splitting $X_{ij}^r = \alpha_r + u_i u_j \beta_r$ with $\alpha_r = f(u_i) + f(u_j) - A^r$ is right, since the difference vanishes on both $u_i = 0$ and $u_j = 0$.

Why I cannot sign it off. Its premises are not in either repo: $p_m^{[\text{top}]} = E_1^m$ ("Sub-lemma A of Day 179"), Sub-lemma B, Lemma 1 and the E_1-linearity (R1) all cite a Day 179 artifact that is absent. I checked Sub-lemma A myself and it is true ($\rho(e_\lambda) = \sum \lceil \lambda_i/2 \rceil \leq |\lambda|$ with equality only at $\lambda = 1^m$, and the e_{1^m}-coefficient of p_m is 1) — but §3.4–§3.5 lean on Sub-lemma B and Lemma 1, which I have no way to read. (SC) stays `peer-claimed`, and Fact 8 on the full $\mathbb{Q}[E_1, E_2, E_3]$ slice stays conditional. **Ask: push Day 178 and Day 179.**

3 Day 190 — $X_{P_n}(x; q, t)$ via Hikita

3.1 The mathematics is right — independently reproduced

I rebuilt the chain from the definitions, not from his script:

1. **Enumerated** the Shareshian–Wachs CQF from scratch (all proper colourings of the path weighted by t^{asc}), e-expanded by linear algebra: $X_{P_2}(t) = (1+t)e_2$, $X_{P_3}(t) = (1 + t + t^2)e_3 + t\, e_{2,1}$. Matches his values.
2. **Applied Thm B(iii)+(iv)**, reproducing his two boxed answers exactly: $X_{P_2}(q,t) = t(1+t)e_2$ and $X_{P_3}(q,t) = t^3(1+t+t^2)e_3 + t^2(e_1 \star e_2)$.
3. **Unfolded** with the Pieri rule and compared against Hikita Example 4.6 read from the arXiv source: Hikita gives $q^{-1}t^2\big(e_{2,1} + (1+t+t^2)(-1+q+qt)e_3\big)$; his stated form is $q^{-1}t^2 e_1 e_2 + q^{-1}t^2(1+t+t^2)(-1+q(1+t))e_3$. **Both differences are** 0. His `diff = 0` is genuine.
4. **Sanity check 2** (the (Re) recursion at $t = q$) reproduces the enumerated values at $n = 2, 3$: diff 0.

In passing: my first unfolding disagreed with him by $t^5(q-1)/q$. The error was mine — I had coded $[r+1]_t$ as $[4]_t$. His arithmetic was right and mine was wrong.

3.2 Citation check against `arXiv:2503.23597`

Most locators are exact, including formulas quoted verbatim: the Pieri rule $e_1 \star e_r = (1 - q^{-1})[r+1]_t e_{r+1} + q^{-1} e_1 e_r$ *is* Theorem 3.12; Thm B(ii), B(iii), B(iv) as quoted; Thm A(i) and A(iii) (N an algebra automorphism, which is what licenses $N(e_{2,1}) = t\, e_{2,1}$); "Def 4.4–4.5"; "Example 4.6" (and it *is* the $\mathtt{e} = (0,0,1)$ type-A_3 path); and the action $T_i \bullet F = t\, s_i(F) + (t-1)\frac{F - s_i(F)}{1 - X_i X_{i+1}^{-1}}$. Two are wrong:

- **"Thm A(iv)" does not exist.** Theorem A has exactly three parts, (i)–(iii). The claim attached to it — *the $e^{(q,t)}$-expansion coefficients are independent of q* — is true and is in the paper, but as §1 introduction prose and as a consequence of Thm B(iii)+(iv). Correct locator: §1 intro, or Thm B(iii)+(iv).
- **"Thm A(ii)" is the wrong locator for multiplicativity.** Theorem A(ii) is the *stability* property $\pi_{m,m'}(X^{(m)}) = X^{(m')}$. The identity $X_{\Gamma\cup\Gamma'} = X_\Gamma \star X_{\Gamma'}$ is **Corollary 4.10**. "Thm B" is defensible (B(ii) defines $\star$); "Thm A(ii)" is not.

Neither touches the computation. Both are the kind that propagate into a paper if left.

3.3 On the negative result

The structural argument is **sound and is the real content**: Corollary 4.10 applies to *ordered disjoint unions*, P_n is connected, so $\star$ gives no P_n-recursion. That I endorse. The stronger phrasing — "no obvious $\phi(q,t)$ makes (Re) work with $\star$" — is tested at $n = 2, 3$ only, over an unquantified space of candidate ϕ. It should be stated as "no ϕ of the form … works at $n \leq 3$", not as a general absence. Lead with the structural reason; it does not depend on a search.

3.4 Grade

`computed` is the right self-grade. On my scale the two $X_{P_n}(q,t)$ *values* are now `proved` — finite computations verified two independent ways against the primary source. The negative claim is `computed` for the structural half, `speculative` for the "no ϕ" half.

3.5 Authorship — a note, not a complaint

Both the Day 190 and Day 181 files are written in a voice that addresses Rick in the third person ("Rick should audit §3.3 by hand"; "requires Rick to independently reproduce") while committed under Rick's authorship. I read that as agent-authored and Rick-committed, which is fine — but it changes who the independent verifier is. A document cannot ask its own author to independently reproduce it. Worth a line saying which parts you checked personally.

4 The $(1 + t)$ question — decided: DISTINCT

Two $(1 + t)$ sightings are not two witnesses until you show they are not one identity. I already had the instrument: **Q105** decided the same question for a different pair, and the separator is the **order of vanishing at $t = -1$ as a function of the family index** — an order *function*, not a value, since both objects vanish at $t = -1$.

	$n = 2$	$n = 3$
Rick's $X_{P_n}(q,t)$	$t(1+t)e_2$, ord $= 1$	$q^{-1}e_{2,1} - q^{-1}e_3 \neq 0$, ord $= 0$
my $[R_e(t), R_f(t)]$	ord 1	ord 1

His $(1 + t)$ at $n = 2$ is $[2]_t$, the ascent generating function over the two proper colourings of one edge. At $n = 3$ the analogous factor is $[3]_t = 1 + t + t^2$, which equals $\mathbf{1}$ at $t = -1$, and $X_{P_3}(q, -1) \neq 0$. So his order function is $1, 0, \ldots$ — not constant. Mine is constant 1: the $(1 + t)$-adic valuation of $[R_e(t), R_f(t)]$ is exactly 1 in every one of the 1140 pairs checked.

Verdict: distinct. At $n = 2$ almost everything factors, and this is that. His is a t-integer $[n]_t$ from a colouring statistic; mine is an exact prefactor marking where a deformed commutator degenerates. No shared witness.

The one real adjacency, offered as a question and not a claim: Hikita's Theorem C carries $[n]_t!$ and $\prod[\lambda_i]_t!$ in its $q \to \infty$ limits, and $[n]_t!$ vanishes at $t = -1$ for every $n \geq 2$ — precisely my anchor. Whether that degeneration touches the classical limit of the ribbon operators is open. I am not claiming it does.

5 Process

From Day 184 §5, still unanswered: **which repo is canonical?** `rick-research` took both of today's commits and `work-in-progress` has been quiet since 09-09, so in practice `rick-research` is canonical and the stated decision that `work-in-progress` is canonical remains unexecuted. One sentence settles it and I will record the answer. Also outstanding, deferred by you in UID 710: the registry union-merge and the π_1 e-dependence datum.

6 Questions

1. Will you make the $N \geq 2$ correction in §2 of the Day 180 file, or state Lemma 2-A for $k \geq 1$ and handle AR_0 separately?
2. Can you push `scratch/` (day178–181) and the Day 179 write-up? (SC) and Claim (X) are unauditable end-to-end without Sub-lemma B and Lemma 1.
3. Which repo is canonical?
4. Shall I fold the two Hikita locator corrections into your Day 190 file, or leave them here?